

KEY LEARNINGS REPORT

Raritone Summer Internship 2026

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Introduction

The Raritone Summer Internship 2026 provided an excellent opportunity to explore Artificial Intelligence, Computer Vision, and Virtual Try-On technologies. Throughout the internship, I worked on research-oriented tasks related to body segmentation, pose estimation, garment mapping, avatar generation, and human body reconstruction. The internship enhanced both my technical knowledge and research skills while exposing me to emerging technologies used in modern fashion technology applications.

Technical Learnings

Understanding Virtual Try-On Systems

One of the major learnings during the internship was understanding how Virtual Try-On systems work. I learned that creating a realistic virtual try-on experience requires multiple technologies working together, including body detection, pose estimation, segmentation, garment fitting, and avatar generation.

I gained knowledge about both 2D and 3D Virtual Try-On approaches and their advantages and limitations.

Pose Detection and Body Segmentation

I learned the importance of body segmentation and pose estimation in Virtual Try-On systems. These technologies help identify body landmarks and body regions that are required for proper garment placement.

Technologies explored included:

- MediaPipe
- OpenPose
- Detectron2

Through this research, I understood how body keypoints are extracted and how they contribute to realistic garment fitting.

3D Avatar Generation

A significant part of the internship involved exploring 3D avatar generation and human body reconstruction.

Technologies studied included:

- SMPL
- PIFuHD
- ICON
- HumanNeRF
- 3D Gaussian Splatting

I learned how modern AI systems can generate realistic digital human representations from images and body information. This helped me understand the future direction of Virtual Try-On systems and digital fashion technologies.

Blender and 3D Asset Exploration

I gained practical exposure to Blender and learned how 3D avatars are visualized and managed.

Activities included:

- Importing avatar models.
- Exploring avatar structures.
- Understanding mesh visualization.
- Performing preliminary garment placement experiments.
- Managing digital assets.

This experience improved my understanding of 3D workflows and avatar-based systems.

Research and Analytical Skills Developed

The internship strengthened my research and analytical abilities.

I learned how to:

- Study technical research topics.
- Compare multiple AI models.
- Analyze advantages and limitations of technologies.
- Create comparison reports.
- Document technical findings.
- Present research outcomes in a structured manner.

These skills are valuable for future academic and professional projects.

Professional Skills Developed

Apart from technical knowledge, the internship helped improve several professional skills.

Skills Improved

- Independent learning
- Technical documentation
- Problem solving
- Critical thinking
- Time management
- Research analysis
- Presentation preparation

The internship encouraged self-learning and helped me become more confident in exploring unfamiliar technologies.

Key Takeaways

Some of the most important takeaways from this internship include:

- Understanding the complete workflow of Virtual Try-On systems.
- Learning how AI and Computer Vision are applied in fashion technology.
- Exploring modern avatar generation techniques.
- Understanding body reconstruction technologies.
- Gaining exposure to Blender and 3D assets.
- Learning the importance of datasets in AI projects.
- Developing technical research and documentation skills.

Overall Experience

The internship was a valuable learning experience that combined research, technical exploration, and documentation activities. It provided insight into emerging technologies that are shaping the future of digital fashion, Virtual Try-On systems, and AI-driven user experiences.

The knowledge gained during this internship has strengthened my interest in Artificial Intelligence, Computer Vision, and emerging technologies, while also improving my confidence in conducting independent technical research.

Conclusion

The Raritone Summer Internship 2026 provided meaningful exposure to AI-powered Virtual Try-On systems and related technologies. Through research, practical exploration, and documentation activities, I gained valuable technical and professional skills that will support my future academic and career growth. The internship served as an excellent opportunity to understand real-world applications of Artificial Intelligence and Computer Vision in modern technology solutions.